

NITIN KUMAR SHUKLA

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EDUCATION

Integrated Master of Technology in Artificial Intelligence

2022 – 2027

Vellore Institute of Technology (VIT)

EXPERIENCE

AI Intern

Factobi | Remote

May 2026 – Present

- Implemented multi-agent workflows using **LangChain** and **LangGraph**, increasing task completion accuracy to **91%** while reducing manual effort by **60%**.
- Architected **RAG** systems using **Pinecone**, **Chroma**, and semantic search, achieving **89% Recall@5** and **93% Top-5 Retrieval Accuracy** on domain-specific document retrieval tasks.
- Delivered and deployed **LLM-powered applications** on **AWS**, **Docker**, and **Kubernetes**; reduced inference latency by **36%** through retrieval optimization, prompt engineering, and efficient deployment pipelines.

Data Science Intern South Eastern Coalfields Limited (SECL), Bilaspur

Oct 2025 – Dec 2025

- Built a healthcare **Document Intelligence** pipeline using **Python**, **OpenCV**, and **Tesseract OCR** for automated claim extraction, validation, and data preprocessing, achieving **94% field extraction accuracy**.
- Analyzed and standardized **500+ healthcare claim documents daily**, performing data quality validation, document verification, and preprocessing on structured and unstructured datasets to support reimbursement analytics and compliance reporting.
- Leveraged **Isolation Forest**-based anomaly detection and statistical analysis to identify billing irregularities and risk indicators, reducing claim-processing turnaround time by **45%** and operational errors by **30%**.

PROJECTS

Automated Audit Verification System

2025

Tech Stack: Python, Whisper, LangChain, FAISS, FastAPI, RAG, RAGAS

- Developed an AI-powered compliance verification platform using **Whisper**, **LangChain**, and **FAISS**, achieving **92% transcription accuracy** while converting unstructured audit conversations into searchable knowledge.
- Constructed **RAG** pipelines with embeddings, semantic search, and LLM-based reasoning, improving retrieval performance by **30%** over keyword-based search; evaluated using **RAGAS** metrics including Context Precision, Context Recall, and Faithfulness.
- Designed **FastAPI** services for transcript processing, entity extraction, and compliance verification, automating **70%** of manual review workflows.

Deepfake Detection Platform

Sep 2024

Tech Stack: PyTorch, ResNet-50, OpenCV, MTCNN, Flask, Docker

- Orchestrated the design and deployment of a **Computer Vision-based Deep Learning system** using **PyTorch**, **ResNet-50**, **OpenCV**, and **MTCNN** to detect GAN-generated media, leveraging transfer learning, face alignment, and image preprocessing for robust classification.
- Trained, tuned, and evaluated **supervised learning models** on the FaceForensics++ dataset, achieving **96.2% accuracy** while utilizing **Grad-CAM-based Explainable AI (XAI)** to interpret predictions and visualize manipulated facial regions.
- Built a **Flask-based inference API** and containerized the solution using **Docker**, reducing deployment setup time by **80%** and enabling reproducible model serving across environments.

TECHNICAL SKILLS

- Languages:** Python, C++, C#, SQL
- AI/ML:** PyTorch, Scikit-learn, XGBoost, Isolation Forest, Hugging Face Transformers, Transfer Learning, Model Evaluation, Explainable AI (XAI), MLflow
- Generative AI & Retrieval:** LangChain, LangGraph, Semantic Kernel, RAG, Prompt Engineering, Fine-Tuning, Multi-Agent Systems, FAISS, Pinecone, Chroma, Embeddings, RAGAS
- Computer Vision & NLP:** OpenCV, MTCNN, ResNet, Grad-CAM, Tesseract OCR, Document Intelligence, Information Extraction, Text Classification, Whisper (ASR)
- Tools & Concepts:** FastAPI, Flask, Docker, Git, GitHub, Statistical Analysis, Data Preprocessing, Feature Engineering.